

Macro Regimes and a Concentrated Growth Fund

Real-Time Evidence on Regime Timing versus Static Diversification, 1990–2026

Khanh Nguyen

Abstract

Allocators often sort months into four macro regimes by the direction of growth and inflation and change their holdings with the regime. We ask whether such a rule can protect a concentrated U.S. growth fund, the JPMorgan Large Cap Growth Fund, without giving up its return. The regimes are built only from data published at the time (766 vintages of industrial production and consumer prices that are never revised), the fund's returns are checked against its annual report, and the overlay is tested under a specification frozen before any backtest. Three results. First, revisions matter: real-time and revised regimes agree in only 81.8% of months, and the usual backtest construction agrees with what was knowable in 67.3%. Second, real-time regimes do not forecast next month's excess return of any of 12 asset classes once 36 tests are accounted for, and for the growth fund the effect of Stagflation reverses sign around 2007 (+14.0% a year relative to other regimes before, −20.9% after). Third, a regime overlay improves the fund's Sharpe ratio from 0.75 to 1.07 in 2007–2026, and its timing component is significant there (+0.22, 90% interval [+0.03, +0.41]); but in 2000–2007 the same timing lowers the Sharpe ratio by −0.18 and deepens the drawdown, for the fund and for ten other equity funds and indices of different styles and regions; these, however, behave like little more than one independent series. A standard predictive regression on the same real-time signals has no out-of-sample skill and adds no value either, and the stock–bond correlation does not explain the change. Holding the overlay's average weights without timing reduces the drawdown in both periods and raises the Sharpe ratio over 2000–2026 (+0.14, 90% interval [+0.05, +0.22]). The robust recommendation is static diversification; regime timing is a bet on the post-2007 era.

1 Introduction

The problem. Many investors hold a single concentrated growth fund. The JPMorgan Large Cap Growth Fund is a good example: a large, well-run fund whose ten largest positions are about half of its assets and which holds almost nothing outside U.S. growth equity. Its returns are excellent in most years, and its drawdowns are deep when growth stocks fall together: −47.6% on month-end data in 2008 (Class I shares), −31.82% from peak to trough in March 2020 on daily data, and −27.81% on month-end data in 2022. The question is whether adding other assets, and changing how much is added with the macro environment, can keep most of the return and cut those drawdowns.

Why macro regimes. The idea that asset returns depend on the macro environment is old and intuitive. Equities do well when growth rises and inflation falls; bonds do well when inflation falls; commodities and gold are expected to help when inflation rises. Classifying each month by whether growth and inflation are accelerating gives four regimes (Goldilocks, Reflation, Disinflation, Stagflation), and a regime overlay holds the assets that historically did best in the current one. The framework is common in practice (Ilmanen, 2011); statistical versions with estimated regime probabilities have a long academic history (Ang and Bekaert, 2002).

Three ways such backtests go wrong. First, macro data are revised. Industrial production is published about two weeks after the month ends and then revised for years, so a backtest built on today’s data uses numbers nobody had (Croushore and Stark, 2001; Orphanides, 2001). Second, return predictability from macro variables is fragile and often vanishes out of sample or across subperiods (Welch and Goyal, 2008). Third, an allocation table chosen after trying many versions on the same data overstates what an investor would have earned (Harvey et al., 2016; Bailey and López de Prado, 2014). A fourth, more mundane, problem is data: mutual-fund price series from free vendors can mishandle distributions, which moves losses between months.

What this paper does. We address each problem directly. Return data are checked against the fund’s annual report to the SEC. Regimes are built from the industrial production vintages published each month and from consumer prices that are never revised, and a look-ahead test with a deliberate leak as a positive control confirms that no decision uses later data. We then ask whether regimes forecast returns at all, and finally test a specific overlay under a specification frozen before any backtest, against six simple alternatives, in an earlier period that played no part in its design, and on ten other equity funds and indices; we also test a standard predictive-regression alternative and one candidate mechanism.

Table 1: Main results at a glance.

Question	Answer
Do revisions change the regimes?	Real-time and revised regimes agree in 81.8% of months
Do real-time regimes forecast next month’s returns?	0 of 36 tests pass a Bonferroni correction
Is the growth fund’s Stagflation effect stable?	+14.0% a year before 2007, −20.9% after
Does the overlay’s <i>timing</i> add value, 2007–2026?	Sharpe +0.22, 90% interval [+0.03, +0.41]
and in 2000–2007?	Sharpe −0.18, 90% interval [−0.32, +0.05]
Does its <i>average allocation</i> add value?	Sharpe +0.09 [−0.01, +0.19] (2007–2026), +0.22 [+0.11, +0.39] (2000–2007)
Same pattern for ten other equity assets?	Yes for all ten, but they behave like 1.2 independent series
Does a predictive regression do better?	No: out-of-sample R^2 −0.23% and −3.31%
Does the stock–bond correlation explain the sign change?	No: within each period both states have the same sign

Contributions.

1. A real-time growth–inflation classifier and a measurement of what revisions change (Section 3).
2. Evidence that regime-conditional returns are not a stable basis for allocation: no asset survives a multiple-testing correction, and for growth equities the sign of the Stagflation effect reverses around 2007 (Section 4).
3. A test of a regime overlay that separates static allocation from timing, uses an out-of-sample period and hold-out assets, measures how independent those hold-outs are, and finds that the static part is robust and the timing part is not (Sections 5–6). A standard predictive regression on the same signals fails in the same way (Section 6.1).

2 Data and verification

Asset returns. Daily total returns from Yahoo Finance, compounded to month-end, for the fund (JLGMX, Class R6, from December 2010; before that its Class I shares SEEGX, the same portfolio with returns 0.31 points a year lower on average because of the higher fee), growth and broad equity benchmarks, three other growth funds, and eight non-equity ETFs. Excess returns are over the 3-month Treasury bill. Before the ETFs exist, long-history proxies are used (Appendix A).

Verification. A single vendor is a single point of failure, so we checked the data against an independent source: the fund’s annual report to the SEC, which states average annual total returns to 30 June 2026 (Table 2).

Table 2: Average annual total returns to 2026-06-30 (%): computed from our data vs. the annual report (or, for IWF, the index it tracks, before its 0.19% fee).

	1 year		5 years		10 years	
	Ours	Report	Ours	Report	Ours	Report
JLGMX (Class R6)	15.53	15.56	12.60	12.66	20.29	20.30
SEEGX (Class I)	15.23	15.26	12.31	12.38	19.98	19.99
IWF vs. Russell 1000 Growth	17.41	17.71	13.51	13.71	18.36	18.58

The fund numbers agree to within 0.07 percentage points a year. Three further checks: adjusted prices equal raw prices plus reinvested distributions on at least 99.7% of days for every series; the two share classes differ by 0.20 to 0.38 points a year, which is their fee difference, and never by more than 0.07 points in a month; and no fund shows a drop of more than 4% on a day when the market moved less than 1.5%, the signature of a missing distribution.

Why this matters: an earlier backtest. An earlier backtest of the overlay studied here, run by the author on vendor monthly prices for 2011–2026, is a useful warning (Table 3). Its compound return for the fund is the same as ours, 16.8%, but its volatility was 4.1 percentage points higher and its losses were placed in the wrong months: -29% in late 2018 that was really -18.5% , and -13% in 2022 that was really -23.4% . The same compound return with more volatility is the signature of distributions recorded in the wrong month: a fund that pays a large capital gain in December shows a fake drop when the payment is missed and a fake recovery when it is caught up. On those data the overlay appeared to raise the return (16.81% against 16.49%); on verified data it lowers it (14.69% against 16.81%).

Macro data. The Real-Time Data Set for Macroeconomists of the Federal Reserve Bank of Philadelphia (Croushore and Stark, 2001) records, for each month from 1962-11 to 2026-08, the history of industrial production as published in the middle of that month: 766 vintages. Each vintage ends with the previous month, except the October and November 2025 vintages, which end in August 2025 because the federal shutdown delayed the release. The last vintage matches today’s FRED series to within 0.33%. For inflation we use the consumer price index *not* seasonally adjusted, which is never revised; its year-on-year change correlates 0.9992 with that of the adjusted index and never differs by more than 0.27 points.

3 Regimes in real time

Timing. The decision is made at the end of month t and applies to returns in month $t + 1$. At that moment industrial production is known through month $t - 1$ (the vintage dated t), and

Table 3: The earlier backtest on vendor monthly prices against verified data, 2011-02 to 2026-05. Sharpe here is (CAGR – T-bill)/volatility, the convention of the earlier backtest. K is the overlay of Section 5; on verified data it uses the real-time classifier.

	Earlier backtest	Verified data
Fund, Oct–Dec 2018	–29.07%	–18.52%
Fund, Feb–Mar 2020	–14.88%	–14.88%
Fund, Jan–Oct 2022	–12.56%	–23.37%
Fund, CAGR 2011-02 to 2026-05	16.81%	16.81%
Fund, volatility	20.73%	16.65%
Fund, Sharpe (old convention)	0.74	0.92
Fund, maximum drawdown	–29.07%	–27.81%
K, CAGR	17.05%	14.69%
K, volatility	15.28%	11.54%
K, Sharpe (old convention)	1.01	1.14
K, maximum drawdown	–13.41%	–17.60%

so are consumer prices.

Signals and regimes. For each series we take the year-on-year growth rate and its change over six months, $d_t = \text{YoY}_{t-1} - \text{YoY}_{t-7}$, computed within the vintage available at t . Year-on-year growth removes seasonality; the six-month change asks whether conditions are improving or worsening. Goldilocks: growth up, inflation down. Reflation: both up. Disinflation: both down. Stagflation: growth down, inflation up.

Variants. The raw classifier uses the signals as they are. A smoothed version averages each signal over three months, and a persistent version changes regime only after a new label has appeared for three consecutive months. For comparison we also build the classifier as it is usually backtested: today’s revised industrial production, seasonally adjusted prices and a fixed lag.

Look-ahead test. We delete every input published after month t , rerun the classifier, and compare every label up to t with the full-data run. At 60 randomly chosen months the labels are identical in 60 of 60 cases. A test that never fails proves little, so we also ran it after letting the next month’s prices leak into the signal: it flagged 60 of 60 cuts.

Table 4: Regime statistics, decisions 1990–2026. Switches: regime changes; spell: average months per regime; last four columns: share of months in each regime.

Classifier	Months	Switches	Spell	Gold.	Refl.	Disinf.	Stagf.
Real time, raw	440	122	3.6	21%	30%	27%	22%
Real time, smoothed	438	94	4.6	20%	30%	28%	21%
Real time, smoothed + persistent	438	58	7.4	20%	32%	28%	20%
Revised, lag 1	440	137	3.2	21%	30%	26%	23%
Revised, lag 2	440	137	3.2	21%	30%	26%	23%
Revised, lag 2, smoothed + persistent	439	57	7.6	18%	30%	32%	20%

What revisions change. Under the same rules and lag, real-time and revised classifiers agree in 81.8% of months. The difference comes from growth: the sign of the growth signal agrees in 83.6% of months (correlation 0.957), that of the inflation signal in 97.7% (correlation 0.997). The usual construction, revised data with a two-month lag, agrees with the real-time labels in

only 67.3% of months: roughly 33% of the labels in such a backtest could not have been known when the trade was made. Smoothing does not wash this out (agreement 70.1%), because the persistence rule turns a small change in timing into a different regime for several months.

Stability. The raw classifier switches 122 times in 440 months, every 3.6 months on average (Table 4); smoothing and persistence reduce this to 58 switches at the cost of reacting later. Delaying the raw classifier by one, two or three further months leaves only 72.3%, 56.1% and 45.2% of labels unchanged.

Table 5: The 2008 crisis month by month. “IP through”: last month of industrial production available at the decision; d^g , d^π : real-time growth and inflation signals (percentage points).

Decision	IP through	d^g	d^π	Real time, raw	Real time, persistent	Revised, lag 2
2007-10	2007-09	−0.22	−0.02	Disinflation	Stagflation	Disinflation
2007-11	2007-10	−0.20	+0.96	Stagflation	Stagflation	Reflation
2007-12	2007-11	+0.26	+1.62	Reflation	Stagflation	Stagflation
2008-01	2007-12	+0.17	+1.39	Reflation	Stagflation	Reflation
2008-02	2008-01	+0.55	+1.92	Reflation	Stagflation	Stagflation
2008-03	2008-02	−0.36	+2.06	Stagflation	Reflation	Stagflation
2008-04	2008-03	−0.74	+1.23	Stagflation	Reflation	Stagflation
2008-05	2008-04	−1.73	+0.40	Stagflation	Reflation	Stagflation
2008-06	2008-05	−2.65	−0.13	Disinflation	Stagflation	Stagflation
2008-07	2008-06	−1.73	+0.94	Stagflation	Stagflation	Disinflation
2008-08	2008-07	−2.73	+1.32	Stagflation	Stagflation	Stagflation
2008-09	2008-08	−3.15	+1.35	Stagflation	Stagflation	Stagflation
2008-10	2008-09	−5.90	+0.96	Stagflation	Stagflation	Stagflation
2008-11	2008-10	−4.39	−0.28	Disinflation	Stagflation	Stagflation
2008-12	2008-11	−5.70	−3.11	Disinflation	Stagflation	Disinflation
2009-01	2008-12	−7.74	−4.93	Disinflation	Stagflation	Disinflation
2009-02	2009-01	−9.32	−5.57	Disinflation	Disinflation	Disinflation
2009-03	2009-02	−9.26	−5.14	Disinflation	Disinflation	Disinflation

An example. In real time the inflation signal stayed positive into the autumn of 2008 because energy prices rose until July, while growth was already falling (Table 5). The raw classifier labels most of 2008 Stagflation and switches to Disinflation in November; the persistent version keeps Stagflation until early 2009; the revised version moves between regimes in a different order. For an allocation rule the differences are large: the same month can call for 25% or 80% in equities depending on which version is used.

4 Do regimes forecast returns?

Test. For each asset we regress the excess return of month $t + 1$ on four dummies for the regime known at the end of month t ; the coefficients are the average excess returns by regime. Standard errors are Newey–West with six lags (Newey and West, 1987). A Wald test asks whether the four means are equal, and a contrast measures Stagflation minus the average of the other regimes, the comparison regime-based de-risking relies on. With 12 assets and three classifiers there are 36 tests, so a Bonferroni correction requires $p < 0.0014$.

No asset survives. With the real-time raw classifier 1 of 12 assets has $p < 0.05$ (high yield, $p = 0.021$), and 0 of the 36 tests pass the Bonferroni level (Table 6). Point estimates are often large: long Treasuries earn +12.7% a year more in Stagflation ($t = 2.01$), and gold earns +12.4% in Stagflation but also +21.4% in Goldilocks. Monthly returns are too noisy, and regime episodes too few, to separate these from chance.

Table 6: Wald p -values for equal mean excess returns across regimes, from the date shown to 2026-09; last two columns: Stagflation minus the other regimes (%/yr) and its t -statistic, real-time raw classifier.

Asset	From	Wald p			Stagflation – others	
		Real time	RT persistent	Revised	%/yr	t
Large Cap Growth (SEEGX)	1992-03	0.501	0.705	0.522	−6.9	−0.90
S&P 500 (SPY)	1993-02	0.590	0.402	0.740	−3.8	−0.54
Russell 1000 Growth (IWF)	2000-06	0.374	0.683	0.217	−9.5	−1.20
Nasdaq-100 (QQQ)	1999-04	0.459	0.739	0.360	−11.6	−1.27
7–10y Treasuries (IEF)	2002-08	0.406	0.985	0.423	+5.1	+1.53
20y+ Treasuries (TLT)	2002-08	0.185	0.932	0.498	+13.2	+1.91
1–3y Treasuries (SHY)	2002-08	0.930	0.276	0.240	+0.5	+0.58
TIPS (TIP)	2004-01	0.605	0.989	0.464	−0.7	−0.21
Gold (GLD)	2004-12	0.262	0.138	0.808	+2.4	+0.33
High yield (HYG)	2007-05	0.021	0.830	0.153	−12.2	−1.65
US dollar (UUP)	2007-04	0.055	0.549	0.588	+4.1	+1.02
Commodities (DBC)	2006-03	0.172	0.487	0.331	−10.8	−0.73

The long-Treasury estimate contradicts a popular rule. Many regime allocations exclude long bonds in Stagflation because of 2022, when stocks and bonds fell together. In real time only 4 return months of 2022 follow a Stagflation label, and in them long Treasuries returned −52% a year. Among years with at least four such months, the four best for long Treasuries were 2020 (4 months, +65%), 2011 (11 months, +37%), 2016 (6 months, +31%), 2008 (7 months, +23%); excluding 2008 they still earn +11.6% a year in Stagflation. A rule drawn from one episode does not describe the regime.

Table 7: The growth fund (SEEGX) by real-time regime: months, annualised excess return (%) and Sharpe ratio, over the full sample and before and after March 2007.

Regime	1992–2026			1992–2007			2007–2026		
	n	Excess	SR	n	Excess	SR	n	Excess	SR
Goldilocks	84	+15.8	+1.05	40	+1.1	+0.08	44	+29.2	+2.05
Reflation	130	+6.4	+0.39	63	+7.2	+0.47	67	+5.6	+0.32
Disinflation	111	+12.2	+0.71	46	−3.2	−0.18	65	+23.0	+1.38
Stagflation	90	+4.6	+0.24	32	+15.8	+1.00	58	−1.6	−0.08
Wald p (equal means)	0.501			0.367			0.004		
Stagflation – others (%/yr)	−6.9 ($t = -0.90$)			+14.0 ($t = 1.36$)			−20.9 ($t = -2.21$)		

The growth fund: the sign changes. Over the full sample the fund earns +4.6% a year above bills after Stagflation months, and the contrast with other regimes is −6.9% ($t = -0.90$). The two halves of the sample disagree (Table 7, Figure 1). Before 2007, Stagflation was the fund’s *best* regime (+15.8%, contrast +14.0%); after 2007 it was close to its worst (−1.6%, contrast −20.9%, $t = -2.21$, Wald $p = 0.004$). After 2007 the regime means also differ significantly for SEEGX, SPY, IWF, QQQ, HYG; before 2007 none of the 4 assets with enough history shows a significant difference. Even after 2007 the result depends on the classifier: with the persistent version the fund’s Wald p is 0.54.

Why the sign might change. Stagflation decision months are spread over 14 calendar years before 2007 and 14 after; the largest clusters are 2005 (7), 1990 (5), 1998 (4), 1992 (3) before and 2011 (11), 2008 (7), 2007 (6), 2013 (5) after. The second half contains the 2008 crash (13 Stagflation months in 2007–2008); the 2022 inflation shock contributes only 3, because year-

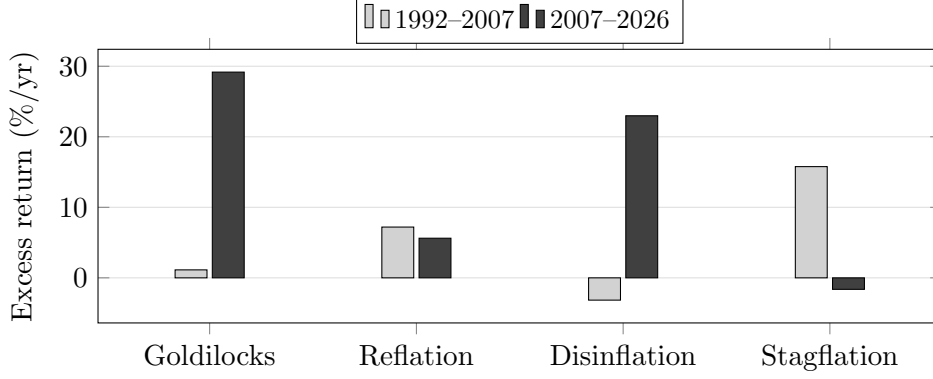


Figure 1: Annualised excess return of the growth fund in the month after each real-time regime.

on-year growth of industrial production was still rising in the decisions from March to August 2022. Campbell et al. (2020) link the change in sign of the stock–bond correlation around 2000 to changes in monetary policy and in the nature of inflation shocks; a similar change could make the equity response to rising inflation depend on the era. This is a hypothesis; the data cannot separate it from the simpler explanation that a few episodes dominate each half.

Testing that mechanism. We can test one version directly. At each decision month we compute the correlation between the fund’s and intermediate Treasuries’ monthly returns over the previous 36 months, which is known at the time, and let the regime effect differ by its sign (decisions from 1995-02). Over the full sample the direction is as the hypothesis predicts: the Stagflation contrast is +2.1% a year when the correlation is positive ($t = 0.16$) and −12.0% when it is negative ($t = -1.25$). But the difference is far from significant ($t = 0.86$), and within each period both states have the same sign: +20.5% and +12.0% before 2007, −10.9% and −24.4% after. The correlation was positive in 48% of months before 2007 and 29% after, so it moves with the era, but it does not explain the sign change: the change follows the calendar, not the correlation state.

Power. The standard error of the fund’s Stagflation contrast is 7.7% a year, so an effect would have to be about 21% a year to be detected with 80% probability at the 5% level (29% and 26% in the two halves). The data do not show that regimes are irrelevant. They show that differences of the size observed cannot be told from chance and do not keep their sign, which is a weak basis for an allocation rule.

5 The overlay and the test design

Table 8: Overlay K: weights (%) by regime, frozen before this study.

Regime (growth, inflation)	Fund	High yield	Gold	Commodities	1–3y Treasuries	Dollar
Goldilocks (up, down)	90	10				
Reflation (up, up)	55	10	30	5		
Disinflation (down, down)	80	10	10			
Stagflation (down, up)	25		25		30	20

The overlay. Overlay K (Table 8) is a typical regime table. It keeps most of the fund in Goldilocks and Disinflation, cuts it deeply in Stagflation, and chooses hedges by regime: gold

and commodities when inflation rises, short Treasuries and dollars in Stagflation, and a high-yield sleeve when growth rises. It was chosen by the author in the earlier backtest of Section 2, on 2011–2026 data, after trying about 40 configurations, and motivated by regime averages and by 2022. It is therefore treated here as a hypothesis, frozen exactly as it was, and tested under conditions it was not designed for.

Specification frozen before any backtest. Every choice below was written down before the first backtest was run.

- **Regimes:** the real-time raw classifier of Section 3; the smoothed persistent and revised versions are sensitivity checks.
- **Rebalancing and costs:** monthly to target weights; turnover measured against the weights after the previous month’s drift; one-way costs of 1–5 basis points per unit of turnover depending on the ETF, zero for the fund.
- **Periods:** 2007-04 to 2026-09, when every ETF exists, containing 2008, 2020 and 2022; and 2000-10 to 2007-03, which contains the dot-com crash and played no part in designing K (sleeves before ETF inception use the proxies of Appendix A; gold futures start in August 2000).

Alternatives. Each answers a specific question.

- **S-K**, the static mix with K’s average weights over the same period: does K’s *timing* add anything beyond its average allocation? Over 2007–2026 these weights are fund 61%, HYG 8%, GLD 18%, DBC 1%, SHY 7%, UUP 5%. S-K is an attribution benchmark, not an investable strategy, because its weights use each period’s regime frequencies. The fund weight is similar in the two periods (61% after 2007, 64% before), so a static mix fixed in advance would have been close to it.
- **G**, generic de-risking: 100% fund in Goldilocks, otherwise 70% fund, 15% short Treasuries and 15% gold. Does choosing different hedges in different regimes matter?
- **60/40** fund and intermediate Treasuries: the textbook balanced portfolio.
- **Volatility target:** fund weight $\min(1, 15\%/\hat{\sigma})$ with $\hat{\sigma}$ the trailing 12-month volatility, the rest in short Treasuries (Moreira and Muir, 2017).
- **10-month trend:** the fund when its total-return index is above its 10-month average, otherwise short Treasuries (Faber, 2007; Moskowitz et al., 2012).
- **90/10** fund and gold: the simplest permanent hedge.

Attribution. $K \text{ minus fund} = (S-K \text{ minus fund}) + (K \text{ minus } S-K)$: what the average allocation contributes plus what timing contributes, applied to the Sharpe ratio, the maximum drawdown and the average return.

Statistics. Differences in Sharpe ratio and drawdown are resampled with the stationary bootstrap (Politis and Romano, 1994) (mean block 12 months, 5,000 draws, all strategies on the same resampled months), a bootstrap version of the test of Ledoit and Wolf (2008).

Success criterion. Regime timing adds value only if the 90% interval of $\text{Sharpe}(K) - \text{Sharpe}(S-K)$ is above zero in both periods, and at least five of six other funds show a positive difference in both periods. The criterion was written after the tests of Section 4, which had shown that the fund’s returns by regime differ before and after 2007; a reader may regard the

earlier period’s result as partly anticipated. Section 6.2 shows that the loss there came mainly from regimes other than Stagflation.

Checks of the code. A 100% fund strategy reproduces the fund’s returns exactly; turnover and costs match a hand computation; K’s weights match the table every month; the volatility and trend weights do not change when later returns are deleted, while a version that peeks one month ahead is flagged by the same test (8 of 18 cuts); 21 of 21 regression checks pass; and two full runs from raw data give identical output.

6 Results

Table 9: 2007-04 to 2026-09, real-time classifier, after costs. Worst 12m: worst 12-month return; Fund wt.: average weight in the fund; Turn.: turnover per year.

Strategy	CAGR	Vol.	Sharpe	Max DD	Worst 12m	Fund wt.	Turn.
K (regime table)	14.3%	11.9%	1.07	−18.2%	−17.6%	0.61	2.94
G (generic de-risking)	13.1%	13.6%	0.87	−33.0%	−28.2%	0.76	1.39
Fund	14.1%	17.7%	0.75	−47.6%	−39.8%	1.00	0.05
S-K (static K weights)	11.4%	11.9%	0.84	−31.0%	−26.6%	0.61	0.32
60/40 fund/IEF	10.0%	10.8%	0.80	−27.0%	−22.3%	0.60	0.30
Volatility target 15%	11.7%	15.1%	0.72	−36.1%	−29.7%	0.87	0.73
10-month trend	11.4%	13.7%	0.76	−19.7%	−18.4%	0.79	3.74
90/10 fund/GLD	13.9%	16.2%	0.80	−42.6%	−35.9%	0.90	0.17

2007–2026 (Table 9). The fund returns 14.1% a year with a volatility of 17.7% and a maximum drawdown of −47.6%, in 2008. K keeps the return (14.3%) with 0.67 times the volatility, and its worst drawdown is −18.2%: its Sharpe ratio is 1.07 against 0.75. The static mix S-K has a similar volatility (11.9%) but a lower return (11.4%) and a much deeper drawdown (−31.0%), so timing explains most of K’s advantage here. The 10-month trend rule matches K’s drawdown (−19.7%) but not its Sharpe ratio (0.76).

Table 10: 2000-10 to 2007-03, real-time classifier, after costs; sleeves before ETF inception use proxies.

Strategy	CAGR	Vol.	Sharpe	Max DD	Worst 12m	Fund wt.	Turn.
K (regime table)	−2.2%	11.9%	−0.35	−47.5%	−36.5%	0.64	4.03
G (generic de-risking)	−2.9%	12.9%	−0.37	−48.2%	−33.8%	0.77	1.25
Fund	−5.2%	16.7%	−0.40	−56.8%	−43.5%	1.00	0.15
S-K (static K weights)	0.2%	10.9%	−0.18	−38.1%	−28.7%	0.64	0.41
60/40 fund/IEF	−0.1%	9.3%	−0.26	−31.3%	−23.8%	0.60	0.41
Volatility target 15%	−0.4%	11.7%	−0.21	−37.3%	−23.9%	0.84	0.60
10-month trend	7.4%	6.2%	0.73	−7.4%	−6.0%	0.47	2.31
90/10 fund/GLD	−3.1%	15.0%	−0.32	−51.8%	−39.5%	0.90	0.27

2000–2007, out of sample (Table 10). The period starts near the top of the technology bubble; the fund loses −5.2% a year with a drawdown of −56.8%. K loses less (−2.2%, drawdown −47.5%), but its static mix does better still (0.2%, drawdown −38.1%): here K’s timing made things worse. The trend rule is the clear winner (7.4% a year, drawdown −7.4%); its average fund weight from October 2000 to December 2002 was 0%, so it sat out the decline.

Table 11: Attribution of K minus fund into static allocation (S-K minus fund) and timing (K minus S-K). Sharpe ratio; maximum drawdown in percentage points (positive = shallower); arithmetic return in points a year.

Classifier	Period	Sharpe		Max DD (pp)		Return (pp/yr)	
		Static	Timing	Static	Timing	Static	Timing
Real time, raw	2007–2026	+0.09	+0.22	+16.5	+12.9	−3.2	+2.6
Real time, raw	2000–2007	+0.22	−0.18	+18.7	−9.4	+4.7	−2.3
Real time, raw	2000–2026	+0.14	+0.09	+20.0	−10.8	−1.1	+1.3
Real time, smoothed + persistent	2007–2026	+0.08	+0.19	+15.2	+19.1	−2.9	+1.6
Real time, smoothed + persistent	2000–2007	+0.24	−0.01	+19.2	−7.0	+4.9	−0.4
Real time, smoothed + persistent	2000–2026	+0.13	+0.11	+18.9	−6.7	−1.0	+1.1
Revised, lag 2	2007–2026	+0.09	+0.30	+16.5	+14.1	−3.2	+3.2
Revised, lag 2	2000–2007	+0.23	−0.04	+20.1	−3.3	+4.9	−0.4
Revised, lag 2	2000–2026	+0.14	+0.20	+20.4	−3.5	−1.2	+2.3

Static allocation versus timing (Table 11). With the real-time classifier, static allocation adds +0.09 to the Sharpe ratio in 2007–2026 and +0.22 in 2000–2007, and makes the drawdown shallower in both (+16.5 and +18.7 points). Timing adds +0.22 in the later period and −0.18 in the earlier one, and changes the drawdown by +12.9 and −9.4 points. Static allocation costs return after 2007 (−3.2 points a year) because it holds less of a fund that did very well; timing gives some of it back (+2.6). The revised-data classifier shows the largest timing gain in 2007–2026 (+0.30): data that were not available flatter the backtest.

Table 12: Differences in Sharpe ratio and maximum drawdown (pp, positive = first strategy shallower), with 90% stationary-bootstrap intervals; $P \leq 0$: share of draws with a Sharpe difference at or below zero.

Period	Difference	Sharpe	90% interval	$P \leq 0$	Max DD	90% interval
2007–2026	K – fund	+0.32	[+0.06, +0.55]	0.02	+29.4	[+7.3, +37.0]
2007–2026	S-K – fund (static)	+0.09	[−0.01, +0.19]	0.08	+16.5	[+7.2, +20.1]
2007–2026	K – S-K (timing)	+0.22	[+0.03, +0.41]	0.03	+12.9	[−1.0, +18.0]
2007–2026	K – trend	+0.31	[+0.04, +0.60]	0.03	+1.6	[−4.5, +13.0]
2000–2007	K – fund	+0.04	[−0.17, +0.40]	0.37	+9.2	[+3.9, +15.6]
2000–2007	S-K – fund (static)	+0.22	[+0.11, +0.39]	0.00	+18.7	[+7.3, +26.2]
2000–2007	K – S-K (timing)	−0.18	[−0.32, +0.05]	0.91	−9.4	[−14.6, −1.1]
2000–2007	K – trend	−1.09	[−2.25, +0.29]	0.91	−40.2	[−58.6, −5.4]
2000–2026	K – fund	+0.23	[+0.04, +0.43]	0.02	+9.2	[+6.5, +31.7]
2000–2026	S-K – fund (static)	+0.14	[+0.05, +0.22]	0.01	+20.0	[+9.9, +26.9]
2000–2026	K – S-K (timing)	+0.09	[−0.07, +0.28]	0.20	−10.8	[−12.2, +13.5]
2000–2026	K – trend	−0.03	[−0.45, +0.38]	0.52	−27.8	[−36.0, +5.5]

Is the timing effect real? (Table 12). In 2007–2026 the timing difference has a 90% interval of [+0.03, +0.41], above zero. In 2000–2007 it is −0.18 with an interval of [−0.32, +0.05], and the drawdown difference [−14.6, −1.1] lies entirely below zero: timing made the worst loss deeper. Over 2000–2026 the interval is [−0.07, +0.28] and includes zero. The static part has intervals of [−0.01, +0.19] (2007–2026) and [+0.11, +0.39] (2000–2007): it is the more reliable part of the overlay. Against the trend rule, K is better in 2007–2026 ([+0.04, +0.60]) and worse in 2000–2007 ([−2.25, +0.29]).

The deflated Sharpe ratio does not answer this. Counting the 40 configurations of the earlier backtest and the 14 run here, K’s deflated Sharpe ratio (Bailey and López de Prado,

2014) is 1.00, but the fund alone also scores 0.97. Every configuration holds the same equity premium, so the statistic only says that the Sharpe ratios are above zero. Whether timing adds value is a question about a difference between strategies, which the bootstrap answers.

Table 13: K and its static mix S-K with the fund replaced by other equity funds and indices, rules unchanged; the last four rows are assets of a different style or region. Timing: Sharpe(K) – Sharpe(S-K); DD: max drawdown of K minus S-K (pp, positive = shallower).

Fund replaced by	2007–2026				2000–2007			
	K	S-K	Timing	DD	K	S-K	Timing	DD
Fidelity Blue Chip Growth	1.07	0.86	+0.21	+6.5	−0.28	−0.11	−0.17	−9.3
T. Rowe Price Blue Chip Growth	1.02	0.76	+0.26	+6.9	−0.10	0.09	−0.19	−9.2
American Funds Growth Fund of America	0.95	0.74	+0.21	+10.1	0.09	0.28	−0.19	−10.8
Nasdaq-100 (QQQ)	1.10	0.92	+0.18	+9.9	−0.31	−0.18	−0.14	−9.4
Russell 1000 Growth (IWF)	1.04	0.83	+0.22	+10.5	−0.32	−0.15	−0.17	−10.2
S&P 500 (SPY)	0.94	0.77	+0.17	+11.8	0.03	0.22	−0.19	−10.1
U.S. large value (VIVAX)	0.77	0.65	+0.12	+10.6	0.26	0.48	−0.23	−8.5
U.S. small cap (NAESX)	0.74	0.57	+0.17	+13.0	0.38	0.57	−0.19	−7.8
International (VGTSX)	0.60	0.40	+0.20	+15.5	0.51	0.62	−0.11	−7.5
Emerging markets (VEIEX)	0.57	0.36	+0.21	+17.8	0.69	0.87	−0.18	−9.0

Other funds and markets (Table 13). For the six U.S. growth and broad-market hold-outs, timing adds +0.17 to +0.26 to the Sharpe ratio in 2007–2026 (positive for 6/6, interval above zero for 4) and is positive for 0/6 in 2000–2007, changing the drawdown by −10.8 to −9.2 points. Four assets of a different style or region (U.S. value, U.S. small cap, international, emerging markets) show the same pattern, 4/4: +0.12 to +0.21 after 2007 and −0.23 to −0.11 before.

How independent is this evidence? Not very. The six original hold-outs correlate 0.92–0.97 with the fund, and their timing series (K minus S-K, month by month) correlate 0.96 with each other on average; the new assets correlate 0.69–0.80 with the fund and their timing series 0.79–0.88. From the eigenvalues of the correlation matrix of timing series, the fund and its six U.S. hold-outs behave like 1.1 independent series and all eleven like 1.2. The hedge legs are identical and equity markets fall together in the same crises, so “the same for every fund” is close to one observation repeated. What the hold-outs do show is that the pattern is not a feature of growth stocks alone; what they cannot do is add independent episodes. Together with the sign change of Section 4, the evidence points to a relation that changed, not to a rule that works.

The criterion fails. Timing is significant in one period and of the opposite sign in the other, for every fund. By the rule fixed before the backtest, regime timing does not add value.

6.1 A standard alternative: continuous signals

K is a hand-made table, so its failure might say little about regime timing in general. We therefore also test a standard approach, fixed before it was run. Each month the fund’s next excess return is forecast by a regression on the two real-time signals themselves (not their signs), fitted on all earlier data from 1992 (first forecast 2000-03). The fund weight is $\min\{1, \max[0.25, 0.60 + (\hat{\mu}_t - \bar{r}_t)/(5\hat{\sigma}_t^2)]\}$, where $\hat{\mu}_t$ is the forecast, $\bar{r}_t$ the historical mean and $\hat{\sigma}_t^2$ the historical variance. The rest is held in a fixed hedge mix taken from K’s table with equal regime weights (HYG 20%, GLD 43%, DBC 3%, SHY 20%, UUP 13%), which uses no data; the static comparison holds 60% fund and the same mix.

Table 14: Continuous-signal model: out-of-sample forecast quality and the allocation built on it, after costs.

	2007–2026	2000–2007
Out-of-sample R^2 vs. historical mean	−0.23%	−3.31%
Clark–West t	0.80	−0.53
Sharpe, model / static	0.89 / 0.84	−0.30 / −0.16
Max drawdown, model / static	−32.2% / −30.6%	−44.4% / −35.7%
Timing (Sharpe difference)	+0.05	−0.14
90% bootstrap interval	[−0.09, +0.20]	[−0.37, +0.01]
Months with the fund weight at a bound	44%	56%

The forecasts have no out-of-sample skill: R^2 is −0.23% and −3.31% against the historical mean, and the Clark–West statistics are 0.80 and −0.53 (Table 14). The allocation’s timing adds +0.05 after 2007, with an interval that includes zero, and −0.14 before. The weight sits at a bound in 44% and 56% of months, a sign that the forecasts are noise relative to the risk scaling; the scaling was fixed in advance and not tuned. The negative result is therefore not specific to the hand-made table: the same real-time signals do not support timing when used in the standard way either.

6.2 Where the timing gain comes from

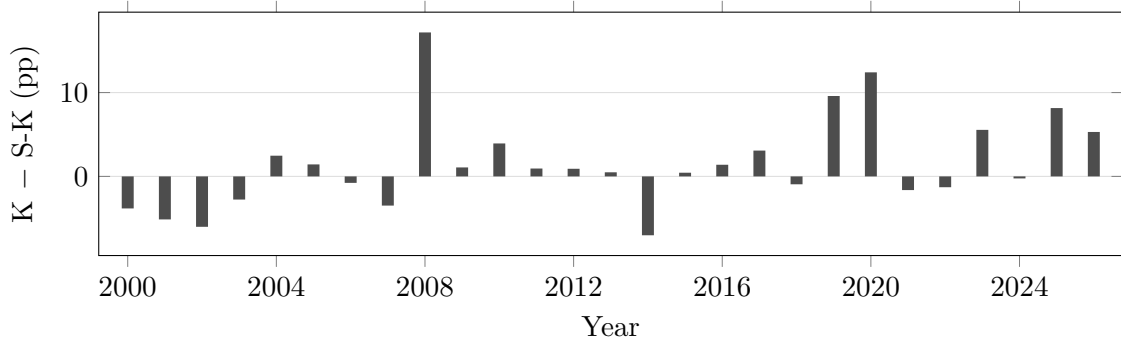


Figure 2: Return of K minus return of its static mix, by calendar year (2000: October–December; 2026: January–September). Exploratory.

The analysis in this subsection was done after the pre-specified tests and is descriptive.

By year. The largest timing gains (Figure 2) are 2008 (+17.2), 2020 (+12.4), 2019 (+9.6), 2025 (+8.2); the largest losses 2014 (−7.0), 2002 (−6.0), 2001 (−5.1), 2000 (−3.8). 2008 alone contributes +17.2 points and 2000–2003 together −17.7. Removing October 2007 to March 2009 from the later period, the Sharpe ratios become 1.22 for K, 1.10 for S-K and 1.03 for the fund: timing still helps (+0.11), by about half as much.

By regime. Summing K minus S-K over the months in each regime (percentage points), the later period gives Goldilocks +17.1, Disinflation +23.6, Stagflation +12.3 and Reflation −2.8; the earlier period gives Goldilocks −12.5, Disinflation −5.9, Stagflation −0.4 and Reflation +3.9. K holds 90% and 80% of the fund in Goldilocks and Disinflation against 61% for its static mix, so most of its timing is a bet that these regimes are good for growth stocks. After 2007 they were (the fund earned +31.1% and +25.1% a year in them); from 2000 to 2007 they were not (−24.3% and −8.1%). Cutting the fund in Stagflation mattered less than the design of K suggests.

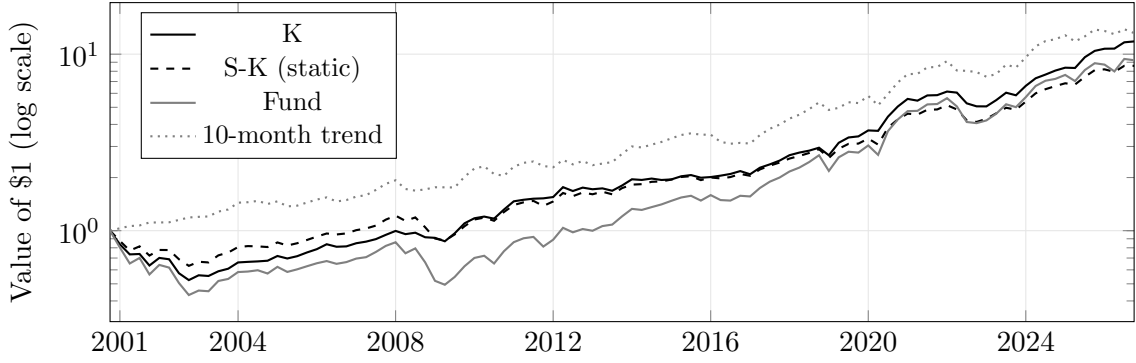


Figure 3: Growth of \$1, 2000-10 to 2026-09, after costs, quarter-end values.

6.3 Costs and implementation

Table 15: Sharpe ratio and cost drag (bps a year) of K at one, two and five times base costs; Sharpe ratio of S-K for comparison.

Costs	K, 2007–2026		K, 2000–2007		S-K Sharpe	
	Sharpe	bps/yr	Sharpe	bps/yr	2007–26	2000–07
×1	1.069	2.8	−0.355	4.0	0.844	−0.180
×2	1.066	5.6	−0.359	8.1	0.844	−0.180
×5	1.059	14.0	−0.369	20.1	0.843	−0.181

ETF trading costs do not decide anything: K turns over 2.94 times its value a year, which costs 2.8 basis points a year at base costs and 14.0 at five times base (Table 15). Two costs not in the table matter more. Mutual funds restrict frequent purchases and redemptions, and moving 30–75% of a portfolio in and out of a fund several times a year may breach those policies. In a taxable account each cut realises capital gains. Both favour the static mix, which only rebalances (turnover 0.32 a year).

7 Implications

For research on regime allocation. Build regimes from real-time data; with revised data a third of the labels in a typical backtest were not knowable, and growth data are the problem. Do not fit allocations to regime means: a table estimated on 2007–2026 would cut growth equity in Stagflation, one estimated on 1992–2007 would raise it. Show that a rule works in a period that was not used to design it, and separate its timing from its average allocation.

For the investor. Holding about 60% of the fund with gold, short Treasuries, high yield and dollars, without timing, reduced the worst drawdown in both periods and raised the Sharpe ratio, significantly before 2007 and over 2000–2026 (after 2007 the interval includes zero); this is the robust part. The timing amounts to holding more growth equity when the classifier says Goldilocks or Disinflation and much less in Stagflation. It paid off after 2007 and cost money in 2000–2007; adopting it is a bet that the post-2007 relation continues. A 10-month trend rule, which needs no macro data, was the best strategy in 2000–2007 and matched the overlay’s drawdown afterwards.

8 Limitations

Few episodes. Each period is dominated by one crash, 2000–2002 and 2008. Bootstrap intervals describe sampling noise around one history, and the hold-out assets behave like about 1.2 independent series; no statistical method creates more crises.

The overlay was designed on 2011–2026. The later period overlaps the data used to choose K, so its results there are partly in sample. Only 2000–2007 is fully out of sample, and it is 78 months long.

Proxies. Before 2007 the sleeves are mutual funds and futures indices. Their tracking errors are small for Treasuries, gold and the dollar but larger for high yield and commodities (Table 16).

One growth series, one horizon, one country. Industrial production covers a shrinking share of the economy; real-time employment or GDP vintages could give a better growth signal. We test next-month returns only, and only in the United States.

9 Conclusion

A growth–inflation classifier run on the data available at the time differs from the backtested version in a third of months and does not forecast next month’s returns of any of twelve asset classes after accounting for multiple tests. For growth equities the effect of Stagflation reverses sign around 2007. A regime overlay for a concentrated growth fund, tested under a frozen specification, improves risk-adjusted returns in 2007–2026 (partly in sample), but its timing hurts in 2000–2007 for the fund and for ten other equity assets, while its average allocation reduces the drawdown in both periods and raises the Sharpe ratio over the whole sample. A standard predictive regression on the same signals has no out-of-sample skill, and the stock–bond correlation does not explain the change. The defensible recommendation is static diversification, possibly with a simple trend rule; regime timing is a view on the current era, not a backtested improvement.

A Proxies before ETF inception

Each ETF is replaced before its first full month by a long-history series. For each pair the version (with or without adding the T-bill rate, for futures-based indices) with the lower tracking error on the 2007–2026 overlap was chosen before any backtest.

Table 16: Proxies compared with the ETF on 2007-05 to 2026-09: correlation of monthly returns, tracking error, and ETF minus proxy average return per year.

ETF	Proxy	Corr.	TE	ETF – proxy
HYG (High yield)	VWEHX	0.906	4.5%	–0.40%
SHY (1–3y Treasuries)	VFISX	0.968	0.5%	–0.11%
IEF (7–10y Treasuries)	VFITX	0.981	2.3%	+0.27%
TLT (20y+ Treasuries)	VUSTX	0.991	2.3%	+0.02%
GLD (Gold)	GC=F	0.992	2.3%	–0.37%
DBC (Commodities)	^SPGSCI + T-bill	0.956	7.3%	–2.27%
UUP (US dollar)	DX-Y.NYB + T-bill	0.994	0.9%	–0.79%

References

- Ang, A. and Bekaert, G. (2002). International asset allocation with regime shifts. *Review of Financial Studies*, 15(4), 1137–1187.
- Bailey, D. H. and López de Prado, M. (2014). The deflated Sharpe ratio: correcting for selection bias, backtest overfitting, and non-normality. *Journal of Portfolio Management*, 40(5), 94–107.
- Campbell, J. Y., Pflueger, C. and Viceira, L. M. (2020). Macroeconomic drivers of bond and equity risks. *Journal of Political Economy*, 128(8), 3148–3185.
- Croushore, D. and Stark, T. (2001). A real-time data set for macroeconomists. *Journal of Econometrics*, 105(1), 111–130.
- Faber, M. T. (2007). A quantitative approach to tactical asset allocation. *Journal of Wealth Management*, 9(4), 69–79.
- Harvey, C. R., Liu, Y. and Zhu, H. (2016). . . and the cross-section of expected returns. *Review of Financial Studies*, 29(1), 5–68.
- Ilmanen, A. (2011). *Expected Returns: An Investor’s Guide to Harvesting Market Rewards*. Wiley.
- Ledoit, O. and Wolf, M. (2008). Robust performance hypothesis testing with the Sharpe ratio. *Journal of Empirical Finance*, 15(5), 850–859.
- Moreira, A. and Muir, T. (2017). Volatility-managed portfolios. *Journal of Finance*, 72(4), 1611–1644.
- Moskowitz, T. J., Ooi, Y. H. and Pedersen, L. H. (2012). Time series momentum. *Journal of Financial Economics*, 104(2), 228–250.
- Newey, W. K. and West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.
- Orphanides, A. (2001). Monetary policy rules based on real-time data. *American Economic Review*, 91(4), 964–985.
- Politis, D. N. and Romano, J. P. (1994). The stationary bootstrap. *Journal of the American Statistical Association*, 89(428), 1303–1313.
- Welch, I. and Goyal, A. (2008). A comprehensive look at the empirical performance of equity premium prediction. *Review of Financial Studies*, 21(4), 1455–1508.